

BRANDON HARGIS

Fayetteville, AR | (469) 401-8762 | Brandonhargiswork@gmail.com
Engineer-in-Training (EIT) Certified | License #9944 | Passed FE Exam

EDUCATION

B.S. Mechanical Engineering — Aerospace Concentration —

May 2025

University of Arkansas, Fayetteville, AR

TECHNICAL SKILLS

Design & CAD:	SolidWorks, AutoCAD, Civil 3D — 3D modeling, assemblies, technical drawings
Analysis:	MATLAB, FEA (ANSYS), XFLR5, OpenVSP, Excel Data Analysis
Manufacturing:	3D Printing (FDM/ABS/ASA-CF), CNC Machining, Laser Cutting, Composite Layup
Testing:	Structural Load Testing, Static G-load Testing, Thrust Stand Testing, Data Acquisition
Documentation:	Technical Reports, ECRs, BOMs, Design Reviews

RELEVANT EXPERIENCE

Mechanical Designer & Test Engineer — Design Build Fly Team —

Aug 2024 – May 2025

University of Arkansas

- Designed and modeled structural systems for a competition UAV in SolidWorks, applying fluid and structural load principles to wing, fuselage, and landing gear components.
- Developed and executed structural test plans using custom-fabricated rigs — including static G-load, tip-to-tip wing, and dynamic drop tests — contributing to an 80% increase in payload carry capacity.
- Collaborated with aerodynamics and propulsion sub-teams to refine designs for durability and mission compliance; prepared technical documentation and contributed to formal design reviews.

Mechanical Engineering Intern — Marshalltown

May 2024 – Aug 2024

- Supported design and development of construction hand tools by creating and revising SolidWorks CAD models and technical drawings.
- Conducted prototype testing and analyzed performance data to validate tool durability and material performance against design specifications.
- Documented engineering change requests (ECRs) maintained accurate BOMs and participated in lean manufacturing and 5S initiatives.

ENGINEERING PROJECTS

Pump & Heat Exchanger Performance Study — University of

Apr – May 2025

Arkansas — Thermal & Fluid Systems Lab

- Applied pump affinity laws to experimentally generate pump and system curves at four operating speeds, achieving 2–4% error vs. theoretical predictions at high RPM.
- Tested a brazed plate heat exchanger in parallel and counter flow configurations, calculating energy transfer rates ($Q = \dot{m}C_p\Delta T$) and effectiveness ($\epsilon = Q/Q_{\max}$) across eight operating conditions.
- Authored a complete technical report including theory, experimental methods, data tables, error analysis, and engineering conclusions.